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India Stack: Digital Public Infrastructure for All

Wealth won't trickle down, so we are raising the floor.

— Nandan Nilekani[1]

Amitabh Kant, India's G20 "Sherpa" during India's G20 presidency, looked up from the report on India Stack, the digital public infrastructure (DPI) built by the Government of India (GoI) to support its sustainable development goals. Nandan Nilekani, co-founder and non-executive chairman of Infosys, was calling. It was December 2023, and Nilekani and Kant had scheduled the call to assess the progress in adopting India Stack. Both were keen to inspire entrepreneurs in India and administrators worldwide to address intractable problems of unequal opportunity using India Stack.

India Stack had been enabled by public investment.[2] Its three layers—identity, payment systems, and data exchange—together provided "online, paperless, cashless, and privacy-respecting digital access to a variety of public and private services."[3] (See **Exhibit 1** for a schematic of the India Stack.) The India Stack accelerated India's start-up ecosystem, which grew from about 900 start-ups in 2016 to about 90,000, including over 93 unicorns, in 2022.[4,5,6] The startups spanned diverse sectors like financial and HR services, health, education, and agriculture. Several rode on elements of the India Stack, especially India's Unique ID (Aadhaar) from the identity layer, DigiLocker from the data layer, and the Unified Payment Interface (UPI), the digital rails for interbank mobile payments layer.[7]

Once a "data-dark" country, India had made impressive progress in Internet consumption, going from 1 GB a month per person in 2016 to 1 GB a day in 2022 and offering one of the lowest-priced data plans in the world.[8] It had the world's highest aggregate mobile data consumption and the second-largest stock of Internet users. It was reported that "three people join the net in India every second, and two are from villages."[9] (See **Exhibit 2** for Internet and Smartphone Users.) Internet access transformed the ease of business and living in urban and rural areas.

Despite the phenomenal growth, in 2023, less than 15% of rural homes had Internet access, compared to 42% of urban households. Digital literacy was low, notably amongst women in low-income segments.[10] In rural India, only one in four women between 15 and 49 years had used the Internet, while almost 49% of males had done so.[11] Historical challenges also persisted: unequal access to quality education, healthcare, and formal employment. There was even debate about whether India

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had moved the needle on poverty since the last official report was in 2011. Some estimated that the number below the poverty line dropped from 22% in 2012 (269 million) to 16% in 2019 (228 million), while others believed it had increased or disappeared.[12,13] [14] Nilekani reflected:

> The digital intensity of society is going to increase. What is the public policy, technology, institutional arrangements, and technology governance in this new world that will create larger value? How do we ensure…technology reaches everyone, and everyone gets access to technology? … technology creates a level playing field that allows innovation to flourish. …that cost is not a barrier to entry? …that it is accessible in multiple languages, so people who don't know English or can't read can still use it? The challenge in digital technology is how we create this in a universal and inclusive way.[15]

Kant's prior stint as CEO of GoI's apex think tank, NITI Aayog, made him familiar with these challenges, but he had seen how DPI could address them. During the first months of the pandemic, it enabled the GoI to make at least one social security payment to the bank accounts of 87% of poor households. While global government spending on IT was close to $600 billion in 2023, "mature" economies accounted for 80% of this.[16] Kant reflected, "The global South can transform the lives of its citizens by adopting India's model of DPI architecture."[17] The GoI was freely sharing its learnings with other countries and signed MOUs with over 20 countries to use elements of India Stack.[18, 19]

Evolution of Digital Public Infrastructure

DPI evolved from government funding. For example, the U.S. government's DARPA (Defense Advanced Research Projects Agency) funded the Internet's inception, Europe-supported CERN developed the World Wide Web, and the National Science Foundation funded the development of the Mosaic browser, the first user-friendly browser at the University of Illinois Urbana Champagne.[20] Private innovations on top of this DPI gave birth to Microsoft Explorer, Amazon, Google, and others. Similarly, the GPS, built by the US Army, enabled private innovation such as "Maps." [21]

Over time, as DPI evolved, the typical approach, perhaps most epitomized by the U.S., was one where the infrastructure of digital platforms was primarily provided by private sector entities, "Big Tech" firms such as Apple, Google, Amazon, and Facebook. These companies used their financial resources and first-mover advantage to exert some control over even open-source technologies such as Linux and Apache. Private players' dominance of DPI gave them access to personal data, enabling personalized advertising and communication and giving rise to "surveillance capitalism." Alarmed by the possibility of ubiquitous monitoring, maybe even the managing of public opinion by big tech, regulators revisited long-standing antitrust doctrine.[22,23] In contrast, the line between privately owned and state-controlled digital infrastructure in China had always been blurry. This increased as the Chinese Communist Party exerted more control, intervening to curtail Alibaba and Tencent, dominant private companies.[24] A third approach was a fully decentralized architecture without regulatory oversight or intermediaries, for peer-to-peer transactions, as seen in cryptocurrencies like Bitcoin. The anonymity, lack of data sharing or dispute settlement guidelines, could allow bad and non-state actors to game the system.[25]

India consciously adopted the route of "the GoI building a public digital highway on which private vehicles could ride." Some of these highways were built in partnership with Nilekani's EkStep Foundation, a non-profit focused on championing digital public ecosystems that could impact billions of people.[26] Similar to public highways, digital highways would not be owned by private entities.[27]

The GoI's approach unleashed creativity in financial inclusion delivery of social services, and consequent ease of living. For example, the average Indian could choose from about 40 instant payment mobile apps. Kant acknowledged, "The partnership of fintech entrepreneurs was key to building a robust digital payments ecosystem in India." Nilekani added, "When you build the right kind of rails at scale and make them available as APIs (Application Programming Interfaces), entrepreneurs can build on that and create multi-billion-dollar valuations."[28]

A Digital "Open, Open, Open" Ecosystem Designed for Inclusion.

The "rails" referred to a technology architecture designed to serve India's 1.4 billion people at minimal cost. In 2015, the GoI launched the Digital India Initiative to create a digitally empowered society and knowledge economy.[29] In 2017, GoI tasked the Digital India Corporation (DIC), incorporated under Section 8 of the Companies Act[a] as a non-profit company under the Ministry of Electronics and Information Technology (MeitY), with fulfilling the goals of Digital India.[30] DIC combined public and private sector input and talent to provide strategic support to ministries and departments at the state and center levels. Abhishek Singh, MD & CEO of DIC, explained that Digital India kept citizens' needs at the center, using scale to reduce unit costs of seamless engagement among government entities and industries. This approach was anchored in simple principles that drove choices. (See **Exhibit 1** again).

India Stack employed a modular approach, using 45 building "blocks," each consisting of APIs, standards, and protocols. Like Lego blocks, each block was independent but could be combined with one or more blocks to craft different solutions. For example, the block that established identity could be built on to authenticate identity or make payments. These blocks could be used by various entities and institutions—government bodies, the private sector, and even other countries. [31]

The protocol approach that enabled "interoperability" was vital. For example, "standard protocols," like the SMTP (simple mail transfer protocol), enabled users to email each other despite being clients of different email providers. In contrast, the platform approach adopted by e-commerce firms such as Amazon tended to become monopolistic, as participants could only interact if they were on the same platform. Another aspect was outsourcing processes to expert third parties.

The mantra was "Open, Open, Open": build and publish on open source and conform to open standards. Mobile phones were the default for the delivery of digital services. Data was an asset. Before data could be accessed, the requester had to notify the citizen, who had to consent to share the data. The guiding principle was to collect the minimum data for a specific purpose. The implementation strategy was to "think big, start small, and scale fast," reuse, and build only in exceptional cases.

Three questions framed citizens' participation in the digital economy: "Who am I? What do I have? What do I do?" These questions were related to interlinked aspects of "digital life," identity, assets, and transactions.

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[a] As per the Companies Act 2013, Section 8 Companies were established to promote non-profit activities, such as education, social welfare, environment preservation, charity. Source: India Filings, https://www.indiafilings.com/section-8-company-registration#:~:text=A%20Section%208%20Company%20is%20a%20type%20of%20corporation%20established,of%20the%20Companies%20Act%202013., accessed January 2024.

Who am I? Establishing, Verifying, and Authenticating Identity

The first building block was related to identity—answering the question—Who am I? Before 2010, most Indians lacked proof of identity, holding none of the conventional forms of identification—a birth certificate, a driver's license, a passport, or a PAN card (Permanent Account Number, proof of being a taxpayer). Proof of address (rent receipts and electricity bills) was out of reach of slum dwellers, precluding most from opening a bank account or buying a mobile phone connection.[32]

Eager to provide all citizens access to these basic services, in 2009, the GoI established the Unique Identification Authority of India (UIDAI) and tasked it with creating a digital identity.[33] Rahul Matthan, partner at Trilegal, a leading law firm, explained, "In a physical world, there are different ways to establish identity and trust in society. In a digital world, you absolutely need to be identified when you are not physically present." The digital identity would be called Aadhaar.

Aadhaar: Digital Identity, Online, Anytime, Anywhere

Aadhaar is the Hindi word for foundation, and the Aadhaar number was the foundation of India's digital infrastructure. It was linked to an individual's biometrics: ten fingerprints and two irises. Biometrics was selected as the only foolproof way to identify more than a billion Indian residents uniquely. Minimal data—name, address, phone number, and date of birth—was collected and stored in a central repository, termed a registry.[34] The principle was of optimal ignorance; the ID system knew no more than was needed to authenticate identity. The Aadhaar authentication process and number did not reveal the location, sex, religion, or ethnicity of a person.[35] This was unique; standard identifiers such as name and mobile number made it easy to surmise these details.

The platform was built for scale - it could enroll 1.5 million people daily at less than $1 per person.[36] By 2014, 600 million people had an Aadhaar ID. The biometric de-duplication and quality checks required for issuance of the digital ID ensured that holders had only one Aadhaar number for their lifetime. The physical Aadhaar card was widely accepted as proof of identity and address.[37] Entities could use Aadhaar only to authenticate identity; they could collect additional data for their businesses if they wished. The authentication process entailed submitting the Aadhaar number by the requester, an OTP (one-time password) driven consent, and transmitting biometrics by the individual to UIDAI, which confirmed only the person's identity with a simple yes/no answer.[38]

In the beginning, enrolling for an Aadhaar number was voluntary. Over time, people who had registered their Aadhaar number with the 'ration shop' (the public distribution system for food grains, sugar, and cooking oil) received their allocated rations swiftly. Rations were not misappropriated if the recipient was physically absent.[b] Similarly, linking Aadhaar numbers to the distribution of liquefied petroleum gas (LPG) cylinders used for cooking stopped out-of-turn hijacking of cylinders. It prevented establishments such as restaurants from obtaining multiple subsidized cylinders against fake identities. Besides making it easier for ordinary citizens to access monthly rations and cooking gas, Aadhar provided a portable identity to several hundred million itinerant workers who moved from rural to urban areas for work.[39,40] Aadhaar linkages saved the GoI $27 billion from losses due to corruption, euphemistically termed 'leakage,' since 2014.[41,42]

[b] Indian labor pools were transient, returning to villages during the harvest season or moving to the plains during the winter.

Verification of Identity: eKYC, Access to Bank Accounts and e-Sign

In 2012, Aadhaar launched the eKYC (Electronic Know Your Customer) service, which instantly verified identity, lowering the cost of onerous paper-based KYC processes previously required to open a bank account. In 2020, Aadhaar launched offline e-KYC. e-KYC was voluntary, but the ease of "anytime, anywhere" helped usage jump from 30 million times in 2015 to over 9 billion by 2021.[43] (See **Exhibit 3** for Aadhaar and eKYC transactions.)

The regulation enabling Aadhaar to be used for eKYC allowed the opening of millions of bank accounts for the unbanked (557 million had no access to bank accounts in 2011), providing an impetus to the GoI's scheme to provide affordable access to financial services to the underserved. [44] The GoI reported that 55% of all bank accounts opened globally between 2014 and 2017 were Indian.[45, 46] By May 2023, the number of accounts had increased to 491 million, with a total balance of $24 billion.[47]

The launch of the Aadhaar-enabled e-Sign in 2015, coupled with Aadhaar authentication and e-KYC, allowed several million Aadhaar holders to sign documents on their devices or an assisted terminal (like kiosks/PoS),[c] lowering acquisition and compliance costs for banks.

Aadhaar as a Foundational ID and Privacy Concerns

Aadhaar evolved from an identity project to an identity behemoth.[48,49] Aadhaar was treated as a foundational identity, a unique identifier used for identity assertion and verification. It was portable across India and enabled access to the accoutrements of daily life - taxes, driver's licenses, school certificates, subsidies, passports, healthcare, banking, pensions, and telephone connections.[50] It differed from functional IDs issued to voters, taxpayers, or employees in the formal sector.

Initially, skeptics questioned the biometric, light data collection approach, and detractors criticized the guardrails around data privacy.[51] In a widely reported incident, Ram Sewak Sharma, founding director general of UIDAI, revealed his Aadhaar number on Twitter, challenging the public, "Show me one concrete example where you can do any harm to me." Several hackers claimed to have found details such as Sharma's phone number, birthday, email, and frequent flyer number, and one person credited one Rupee to Sharma's account. UIDAI described these as a 'cheap publicity stunt,' explaining that since Sharma was a government official, these details were available in the public domain. UIDAI confirmed that no hacker had accessed any data from the Aadhaar database.[52,53]

Societal and legal contestations ensued, culminating in a 2018 Supreme Court judgment that explicitly added the right to privacy to the fundamental rights guaranteed by the Indian Constitution, protected as an intrinsic part of the right to life and personal liberty.[54] It held that broadly, the benefits of Aadhaar outweighed its risks, allowing Aadhaar to be widely used to establish identity. However, it was not required to get a mobile connection or a bank account or for exams to enter the elite Indian Administrative Service.[55] Aadhaar cards were issued with all but the last four digits masked. An app, mAadhaar, was introduced, allowing people to access a digital copy of their Aadhaar cards, equivalent to a physical identity card. However, complaints that privacy was an afterthought continued.

Critics worried that the centralization of the identities of 99% of Indian residents would make it vulnerable to attacks. Experts refuted this as UIDAI did not store information beyond the basic details and Aadhaar number. When an individual gave their Aadhaar details to a bank for verification, UIDAI

[c] PoS is the point of sale where a customer makes payment, such as the payment counter, self-payment kiosk, or mobile PoS.

would respond to the bank with Yes or No. UIDAI did not store any more information, such as bank account details, health records, or education, in its database.

In 2018, UIDAI introduced tokenization.[56] A token or virtual ID was temporary and revocable, limiting the ability to engage in deep profiling, that is, identifying a person from the pattern of his or her aggregate digital activity. Users could change their virtual ID or generate a new ID for each new service they accessed. [57] Still, many noted that as Aadhaar scaled, it would struggle with security and privacy on one side and scale and utility on the other.[58]

What do I have? Owning, Accessing, and Sharing Data

The policy framework proposed by NITI Aayog for data governance, Data Empowerment, and Protection Architecture (DEPA) recognized that using digital infrastructure created digital trails.[59] DEPA, explained Kant, aimed to empower individuals to "use their digital trails for their benefit," thus addressing the risks of a data monopoly. Nilekani added, "Data accumulates in companies or governments. Digital activity creates a digital exhaust, which aggregates somewhere."

DEPA had three pillars. First, it was based on fairness principles enshrined in legislation such as the European GDPR. These principles included consent, minimalism, and data restrictions, giving individuals control over their data. DEPA recognized that Indian residents owned their data and had to consent to share it explicitly. Second, a category of regulated entities was created as "traffic cops" to facilitate the sharing and consumption of data, giving citizens control of the data and their digital trails. Kant added, "DEPA inverts the traditional Western model where data is simply used to advertise and sell products, to one where data can be used to empower a billion Indians." Third, while a privacy law had yet to be enacted,[60] Matthan explained that India's "techno-legal" approach differed from the U.S., where private players set the rules, or the European approach, which was regulation-heavy and slower in implementation. India embedded the desired compliant behaviors, such as mandatory user consent for data access, into the design of the technology so that just using Indiastack ensured compliance.[61] Regulation was, therefore, not left to private entities and did not lag behind fast-moving technology. Other organizations, such as the iSPIRT Foundation (a think-tank), supported the implementation of this framework through supportive and advisory roles.[62,63]

In addition to creating a policy framework for sharing data, building blocks were developed to enable data sharing. One such initiative was DigiLocker. [64] DigiLocker empowered users with control over their credentials by answering the question – "What do I have?"– and enabling them to share this quickly and safely. DigiLocker users could access digital versions of documents such as driver's licenses and academic certificates from the original issuer and consent to share them securely with a prospective bank or an employer, who could verify the documents in real time.

DigiLocker had only 1.2 million users by 2016. This increased once the Information Technology Rules 2016 Act allowed documents issued in the DigiLocker system to be acceptable alternatives to original physical documents and introduced a facility for self-e-signing legacy documents.[65] In February 2017, a bank began providing access to documents in DigiLocker from within its net banking application, allowing users to e-sign and forward them, eliminating the inconvenience of requesting the bank for an attested statement. Over time, other banks, municipalities, school systems, and hospitals became issuers, and organizations of all hues and sizes became users.[66] One user posted, "I could pull my docs, like my college mark sheets, driver's license, PAN card, etc., in a few seconds just by providing one or two details.[67] By 2023, DigiLocker had over 166 issuers, 160 million users, and had issued 5.6 billion documents.[68] Pramod Varma, chief architect of Aadhaar and co-chair at the Center of

Digital Public Infrastructure, said, "Verifiable credentials are the only way India will move from a high cost, low trust environment to a low cost, high trust transaction environment."

Many services were built using DigiLocker. Varma pointed to DigiYatra, a digital security screening system introduced by the Ministry of Civil Aviation to resolve citizens' complaints regarding delays entering airports. DigiYatra used DigiLocker to verify the individual's Aadhaar ID when they signed up. One passenger exclaimed, "I reached Delhi airport late but caught my flight as DigiYatra whizzed me through the queues."[69] Varma said, "This exemplified the birth of new institutional arrangements between the state and the private sector in the provision of civic services.[70]

What do I do? Transacting Easily and Seamlessly

With the data infrastructure in place, DIC focused on shepherding building blocks that enabled digital access to various public and private services.

UPI: Immediate Peer to Peer (P2P) and Peer to Merchant (P2M) Payments

A seminal block was the Unified Payment Interface (UPI), launched in April 2016 by the National Payments Corporation of India (NPCI), a nonprofit entity owned by 56 member banks, later increased to 67, with public sector banks forming 45% of its shareholders.[71, 72] UPI enabled smartphone-based instant money transfer across different banks or payment wallets. (See **Exhibit 4** for the UPI payment process and **Exhibit 5** for UPI at small stall owner.) While fintech companies built apps to facilitate UPI payments, NPCI launched a free reference app called BHIM (Bharat[q] Interface for Money) to gain momentum. Adoption was initially slow but grew in the following years with increasing Internet access and GoI efforts to discourage cash transactions, which were expensive to manage. A 2016 report found that the cost of cash transactions in India was equivalent to 1.7% of its GDP.[73]

The GoI eliminated UPI transaction fees, drawing from its experiences with credit card usage when businesses refused credit cards for low value transactions. The vision was to allow the benefits of digital payments to be accessible to all.

UPI's launch coincided with the launch of Reliance Jio's 4G network in India. Reliance Industries Limited, one of India's biggest conglomerates, invested tens of billions of dollars into Jio, offering cheap mobile phones, free "voice calls," and 4GB of free data daily. [74] The uptake, in a country where only 9% of rural areas had Internet access, was instantaneous, especially as Jio used Aadhaar eKYC to issue an active connection in minutes.[75] 100 million enrolled in six months.[76, 77] Reliance Jio's most extensive base of tens of millions of customers paying $2.5 monthly was in the populous state of Bihar, which had the country's lowest literacy levels.

Vikas Choudhury, President of Reliance Jio, who was involved in the network rollout, shared that in this market, where most customers spoke Bhojpuri (an Indian dialect), little Hindi, and no English, a Hollywood film in English was the most viewed item! He said, "The underprivileged are virgin markets with aspirations. Once you give them data, they will consume it. The Jio network and phone helped millions become entrepreneurs. One young adult hooked up his phone to two loudspeakers, created playlists, and became the village Disc Jockey. He charged a small fee for his musical events and eventually got employed across a cluster of villages."[78]

With Jio leading the way, telcos reduced the price of mobile data plans, from 1 GB of data costing roughly $3 in 2015 to less than a cent in 2021. Affordable data and phones for as little as $12 led users to access the Internet through mobile phones, making India a 'mobile first' country. Within six months

of Reliance Jio's launch, India had one of the highest numbers of mobile data users worldwide.[79] By 2019, across all telecom providers, ~637 million residents had access to the Internet.[80] (See **Exhibit 2**)

The Internet revolution, GoI efforts to discourage cash payments, NITI Aayog-led efforts in partnership with NPCI, and the Covid-19 pandemic led digital payments to skyrocket.[81] The ratio of digital transactions to cash rose for peer-to-peer (P2P) and peer-to-merchant (P2M). By 2023, over 445 banks and 68 applications were on UPI.[82] (See **Exhibit 6a** and **6b** for UPI Statistics.) Private players such as PhonePe, GooglePay, Paytm, and Amazon Pay innovated on the platform, offering payments, loans, and investments.[83] UPI was the 5th most extensive payment system in the world, behind only Visa, AliPay, WeChat-pay, and Mastercard, by volume, with over 7.5 billion payments in February 2023.[84,85] About 80% of these were on PhonePe and Google Pay, with Paytm in third place.[86] By 2023, UPI's 8.6 billion transactions per month overshadowed the transactions done on India's 6 million PoS machines, demonstrating the power of interoperability to catapult the speed of acceptance.[87]

The financial viability of UPI was debated. Some argued that UPI's biggest impact was "when consumers could choose the product that best suited their preferences."[88] Still, there was no revenue for the key players, unlike for the providers of other public infrastructure.[89] Public highways, for example, generated $4 billion in 2022 through a digital toll collection system with 96.34% penetration. Others argued that the GoI used regulatory clout to tilt the playing field in favor of UPI by making it free, thus eroding fee-based transactions delivered by Visa and Mastercard.

Moreover, in recent years, questions about NPCI emerged. Incorporated in 2008 as a non-profit under Section 8, NPCI had launched and operated many retail payment systems, including IMPS, the widely used real-time interbank funds transfer system, Aadhaar-enabled payment service, the RuPay credit card system, and FASTag, an electronic toll collection system.[90] It was featured in the TIME 100 Most Influential Companies in 2023. In FY23, its $99.6 million net profit made it one of India's most profitable fintechs.[91] Its subsidiary, NPCI International Payments Ltd (NIPL), was responsible for formulating business strategy and deploying NPCI's technology internationally.[92] NIPL targeted countries such as Belgium, The Netherlands, and Luxembourg.[93] Some questioned whether NPCI was wielding too much power and if this meant it should give up its 'non-profit' tag.[94]

Despite the hurdles, UPI's success underscored the transformative potential of open protocols, and led the Bank for International Settlements – an umbrella organization owned by central banks – to issue a working paper co-authored with Nilekani that sought to redefine the core of financial markets using digital technologies. It introduced the concept of 'Finternet,' envisioning a futuristic financial system where technology would weave together multiple ecosystems. By harnessing the power of digital IDs, unified ledgers, and tokenization, users would be able to seamlessly transact across wide arrays of asset classes—from cash, stocks, and bonds to more complex holdings like real estate, jewelry, and artwork. In theory, this would herald an era of instant asset transfers, eliminating the need for traditional verification processes and improving efficiencies in the financial system.

Innovations at Population Scale for Financial Services

With digital payments creating digital trails, further innovations in the financial arena followed, as entrepreneurs took advantage of new regulations and building blocks to fill voids that hindered business.[95] Kant emphasized, "Start-ups focused on personal finance such as money management, investments, trading, and insurance cover." For example, in December 2016, the Reserve Bank of India (RBI) permitted financial institutions to deliver unsecured loans up to $730 through mobile phones without physical verification and paper proof of identity and address.[96] Bajaj Finance, an incumbent with a $40 billion market cap, leveraged this to grow dramatically. Many fintech apps embedded the

GoI-service Bharat Bill Pay, a utility bill payment network, with UPI, enabling bill payments with one click.

Microfinance institutions found that since 25% of consumers in lower income groups were "likely to opt for digital modes of transactions," they could use DPI for onboarding, disbursements, and collections.[97] Over 95% of the bottom 40% income group had a bank account and debit card, thus reducing expenses and bolstering the growth of the microfinance sector.[98]

Access to Education and Healthcare

Apart from financial services, the GoI focused on creating DPI to provide quality education at scale. India had about 300 million students in schools and 100 million in colleges and universities. With the combined customer base of digital education solutions developed by start-ups and non-profits being limited to about 16 million, the GoI launched free open education platforms such as SWAYAM (Study Webs of Active Learning for Young Aspiring Minds), which saw ~25 million students enroll, and DIKSHA (Digital Infrastructure for Knowledge Sharing), built on earlier DPI blocks for school education. (See **Exhibit 7** for illustrative examples in EdTech and HealthTech.)

With the onset of Covid-19, the GoI built digital solutions using Aadhaar and DigiLocker in healthcare. In January 2022, Sharma of UIDAI emerged from retirement and, along with his team of five members, deployed CoWIN in record time to drive Covid-19 vaccination.

India had a well-established pediatric vaccination program that reached 25 million children yearly.[99] Still, Sharma recognized the challenges of a vaccine rollout for nearly a billion adults, many digitally unsophisticated. Compounding issues of scale was the need to minimize vaccine wastage and prioritize frontline workers, older people, and ones with co-morbidities. The CoWIN platform was the answer that evolved to meet the challenges that arose. [100] A person could enroll up to five others and choose different vaccination centers for each shot. This helped transient workers and those who lacked digital access. CoWIN supported voluminous traffic, including 25 million vaccinated in one day, and eventually recorded a billion hits in less than a year.[101]

CoWIN issued digital vaccination certificates linked to Aadhaar IDs and passports that could be authenticated using the QR code, popularizing tamper-proof "digital credentials." Sharma shared, "CoWIN is now being repurposed for other healthcare initiatives, such as blood and organ donation. The vaccination module is being converted into a Health Management Information System (HMIS)…which small clinics can use."[102] Kant added, "GoI-supported initiatives have created over 460 million health accounts to enable seamless storage and retrieval of medical records. Eventually, the health and finance sectors could collaborate to ease expenses and create a new future for healthcare."[103]

Internet & E-Commerce for All

The GoI tried to democratize access to e-commerce. Nilekani explained, "When designing something for a billion people, you have to think differently, so how do you make it accessible to everyone?"[104] This approach differed from that of private players, who usually began by looking at the top of the market- the most affluent few tens of millions. Matthan of Trilegal added that it was not just about providing access but empowering all.

Despite increased e-commerce volumes, penetration of sellers remained in single-digit percentages, confined to large urban centers.[105] In response, the GoI announced the Open Network for Digital Commerce (ONDC), a company with both GoI and private funding, in December 2021. ONDC sought to overcome the restrictions of e-commerce platforms such as Amazon, which charged huge

platform fees, only allowed registered buyers and sellers to transact, and was chiefly confined to large urban centers. Shireesh Joshi, the chief business officer of ONDC, elaborated:

> Our goal is to increase e-commerce penetration dramatically to population scale. For conducting e-commerce via platforms, the owner of a taxi or a restaurant can only access a couple of players with scale in terms of riders or customers. Sellers are subject to commercial terms that can be unviable, like high commissions, long settlement periods, or costs of return, despite sending the correct product to the customer on time. These limitations of a platform-based approach can only be solved by a network approach like with UPI.[106]

ONDC was a network that allowed buyers and sellers on different apps to communicate with each other.[107] When a consumer searched for an item, the search query was relayed to the gateway, which explored the ONDC registry and shared the search query with all relevant seller apps. Anand Mahindra, HBS MBA 1981, chairperson of Mahindra and Mahindra, tweeted, "We're on the brink of a world-beating retail revolution. This will democratize e-commerce, liberating us from the dominance of a few giants (and) empowering millions of small shop owners."[108] ONDC used the unbundling principle so sellers could fulfill the order themselves or use a logistics partner via ONDC's network. A collective of small farmers shared that ONDC gave "nationwide visibility to local treasures of rice, millet, and honey, creating a farm-to-fork value chain."[109]

While e-commerce was a higher-margin business than payments, Sameer Nigam, co-founder of PhonePe, believed the challenge of digitizing millions of firms would make it slow to scale.[110] Skepticism around ONDC surfaced as the system was more complex as the transaction loop required offline fulfillment and enablers like grievance redressal and returns. The stakeholders in ONDC were stores and small businesses that might not provide the same level of financial support as fintech platforms did for UPI. ONDC started as a platform for small businesses, but as more big players were invited, it raised concerns about unequal competitiveness.[111] Local stores or small businesses could still struggle to compete with larger players' discounts and sales. ONDC did not yet have a mechanism to clamp down on anti-competitive activities. Nigam said, "It (e-commerce) is somewhere between a moonshot – because of ONDC – and a huge market to tap into… The opportunity is larger than payments, but it is much riskier…nowhere in the world has any government or industry aggregated so many aspects of fulfillment – logistics, inventory management, seller, and buyer platforms..."[112]

To increase the adoption of the Internet in India, it was important to overcome the language barriers. In 2022, the GoI launched Bhashini (BHASHa INterface for India), an AI-powered language translation platform, making the country's digital initiatives available in 22 Indian languages. It aimed to substantially increase the content in Indian languages on the Internet. Users could access their bank accounts, transfer money, book LPG cylinders used for cooking, and renew insurance in their language. The goal was to facilitate digital inclusion by leveraging AI/NLP and transcending language barriers on the Internet.[113] Bhashini introduced *Bhasha Daan*, an initiative to gather Indian language inputs, encouraging individuals to contribute anonymously, enriching datasets for AI model training. [114]

Addressing Businesses' Needs: Credit and Identity

Another pillar was the OCEN (Open Credit Enablement Network) to provide equal access to credit. It was designed by the iSPIRT Foundation to make lending as simple as sending a payment over UPI.[115] While banks and non-banking financial companies (NBFCs) lent to large enterprises, microfinance institutions and moneylenders served the bottom of the pyramid inefficiently and expensively. The

result, by some estimates, was about $530 billion of unfulfilled demand for credit since only 14% of MSMEs had access to formal credit.[116] OCEN was "a tool for the missing middle." Mathan explained:

> No one is willing to lend to a streetside vegetable vendor, who every morning takes a loan at exorbitant rates from a money lender to buy vegetables from the market, repaying the money lender from his earnings at night. He doesn't have collateral but has used UPI for five years, which shows that, for those five, he went to the market daily, took Rs.20,000 as his initial loan, and was able to pay back Rs.22,000 with interest. Can he use that trail to get a loan? If the vegetable seller goes to the bank to get a statement, it's one day's earnings gone. Often, you have to go from lender to lender. OCEN can change all this.

Approximately 11 million merchants were paying goods and services taxes, generating government-certified invoices that could be used as "information collateral" to build creditworthiness.

While the concept of open finance was not new, DEPA implemented it differently. It introduced a new category of regulated entities, called Account Aggregators (AAs), that facilitated the 'secure and easy sharing of data with user consent.' Merchants could access credit by authorizing their banks to share their financial information with prospective lenders through an AA which also acted as a consent manager. Matthan offered a hypothetical example, "This could extend beyond financial history. For example, if you are a columnist for The New York Times and The Atlantic wishes to engage you, but first wants The New York Times to confirm that you have written the articles you say you have. The New York Times could use consent managers to provide authenticated documents, effectively certifying that this was written by you." (See **Exhibit 8** for Consent Manager.) DEPA offered a privacy double-blind and would keep minimal data for a specified time. Matthan explained,

> The consent manager knows the data provider and consumer but doesn't know what the data is. The data consumer knows what the data is but does not know who the data provider is. The data provider knows what the data is but does not know who the data consumer is. So, no one has all the information, and privacy is preserved.

Many hoped that this would spur a shift from traditional asset-based lending to cash flow-based lending based on income, receipts, and tax returns. Small business owners and others who relied on savings or moneylenders would be incentivized to join the formal economy if they could access short-tenure, low-cost credit from the formal system at better terms.[117] Nilekani said, "This combination of credit with UPI, enabled by the AA system and ONDC, will fundamentally reorder India's supply chain. India is showing the world how to combine regulation and innovation."[118]

In addition to the credit and visibility, new companies emerged and built solutions based on IndiaStack elements to address other pain points businesses faced. One example was Wify, a B2B services provider for the installation or assembly of furniture.[119] Wify used IndiaStack elements to authenticate the identity of people who joined the platform, making its B2B customers' end consumers more comfortable booking a service through the company.

Sharing Know-how with the Global South

DPI's potential to reach more citizens led many countries to ramp up their national identity program efforts. Malawi's national ID system reached 9 million in three years, and Cambodia's payments system served 6 million. CV Madhukar, CEO of Co-Develop, a global not-for-profit fund (with partners such as Bill and Melinda Gates Foundation, Nilekani Philanthropies, the Rockefeller Foundation, and Omidyar Network), to help countries build DPI, shared:

> After the pandemic, over 50 countries reached out to the World Bank and other institutions for support to build these systems for their citizens. Understandably, countries feel a sense of urgency to develop these capabilities, but they must ensure that these systems are inclusive, safe, and equitable. Accomplishing this will require reimagination and adopting an infrastructure mindset; DPI requires deep conviction and not deep pockets.[120]

Countries in the global South expressed interest in different aspects of the India Stack. In 2018, India made know-how on national ID systems available to governments through an open-source Digital Public Good – MOSIP (Modular Open-Source Identity Platform).[121] By February 2023, MOSIP was in various stages of adoption in Burkina Faso, Ethiopia, the Philippines, Sierra Leone, Sri Lanka, and Togo.[122] In 2023, India and Singapore agreed to link UPI and Singapore's PayNow system, India's first collaboration with a P2P payment facility in another country. India also finalized agreements with Bhutan, Nepal, Oman, UAE, France, Malaysia, Japan, and the U.K. to adopt UPI.[123]

In 2023, at the G20 summit in Delhi, India announced a plan to build a Global Digital Public Infrastructure Repository (GDPIR), a virtual repository of DPI voluntarily shared by G20 members and beyond. This would be particularly helpful for countries with small populations or those that especially lacked digitally skilled workforces and when public sector workforces were aging. For smaller countries, this meant lower implementation costs and faster adoption of digital services."[124]

Other countries tweaked their offerings. Brazil's central bank launched an instant payment system, Pix, in 2020 which quickly became the country's most popular payment option. Brazil addressed the financial viability of Pix by charging merchants fees determined by the market, while keeping it free for users.[125] In 2022, despite Brazil's smaller population (less than a sixth of India's) Pix's user base was about half of UPI's.[126] Although the Pix protocol bore similarities to UPI, it restricted participation to the banking sector, while UPI allowed tech companies to enter the ecosystem, which catalyzed innovation from non-banking sectors, evidenced by prominent payments applications like PhonePe and Google's GPay, which were spearheaded by technology-first companies.

The Way Forward

Nilekani and Kant discussed how India had charted a path to digital and financial inclusion, with the private sector innovating on top of India Stack. The layers—establishing who I am, giving me control of what I have, and enabling interoperable transactions—could be game changers in creating digital access to public goods. Nilekani mused:

> An Indian in Bihar using a smartphone will have the same digital footprint as a person in Boston because both are doing similar activities. So, people's digital footprint will be as high, but their income will not. If data is going to be the new collateral for people to improve their lives, if I'm going to create a digital exhaust of my life, can I use that data to get access to credit, healthcare, jobs, and skills… this is one of the most profound ideas -- how do we flip data around to make it useful to people, to empower them.

The architects of India Stack had embraced the open network approach, foreseeing that as data trails increased, it would be critical to prevent this data from being captured by rent-seeking platforms.

Despite the challenges, Kant and Nilekani believed that India's approach to DPI provided a tremendous opportunity. Kant believed that DPI was "an idea whose time has come and is here to stay."[127,128] He said, "India's DPI has given it a head start in the digital era." He felt that the India Stack

approach of public rails over which private innovation could thrive could create an inclusive economy and an equitable society. At the same time, both were aware that some saw the proselytizing of DPI as India's "belt and road initiative" to increase India's soft power. However, they believed that a generous approach to sharing open-source software would serve the world well.

Exhibit 1a India Stack: Select Illustrative Examples

What do I do?	**Democratizing access to key services across payments, education, health, e-commerce and credit via interoprability**				
	Enabling Interoperable, Instant and cheap payments UPI (Unified Payments System) BharatPay	**Enabling credit flow between borrowers, lenders, and credit distributors** OCEN (Open Credit Enablement Network)	**Enabling an inclusive e-commerce ecosystem** ONDC (Open Network for Digital Commerce)	**Enabling digital healthcare for all** CoWIN (COVID Vaccine Intelligence Network)	**Enabling digital education for all** DIKSHA (Digital Infrastructure for Knowledge Sharing)
What do I have?	**Empowering residents to access, control and securely share data**				
	DigiLocker - Document/Credential infrastructure Account Aggregator - Consent mechanism to share information across financial institutions				
Who am I?	**Giving every resident a unique ID and enabling them to prove "I am who I claim to be"**				
	Aadhaar - Unique ID eSign - Electronic – Know Your Customer eKYC - Digital Signature App				

Source: Compiled by casewriters from National Payments Corporation of India, "About Us," https://www.npci.org.in/who-we-are/about-us, accessed June 2023; Goods and Services Tax, "About Us," https://www.gst.gov.in/about/gst/history, accessed June 2023.; Bharat Bill Pay, "About Us," https://www.bharatbillpay.com/about-us, accessed June 2023. ; DigiLocker, "About Us," https://www.digilocker.gov.in/about/about-digilocker, accessed June 2023.; India Stack, "Data," https://indiastack.org/data.html, accessed June 2023; "Know all about Account Aggregator Network- a financial data-sharing system," Ministry of Finance press release, September 10, 2021, on PIB website, https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1753713, accessed June 2023.; PhonePe gets in-principle approval as an Account Aggregator from RBI," PhonePe press release, August 26th, 2021, on PhonePe website, https://www.phonepe.com/press/phonepe-gets-in-principle-approval-as-an-account-aggregator-from-rbi/, accessed June 2023; Open Network for Digital Commerce, "About Us," https://ondc.org/about-ondc/, accessed June 2023.; "At 87%, India has the highest FinTech adoption rate in the world against the global average of 64%": Shri Piyush Goyal," Ministry of Commerce & Industry press release, September 30, 2021, on PIB website, https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1759602, accessed June 2023; CoWIN, "Home," https://www.cowin.gov.in/, accessed June 2023; Unique Identification Authority of India, "About your Aadhaar," https://uidai.gov.in/en/my-aadhaar/about-your-aadhaar.html, accessed June 2023.

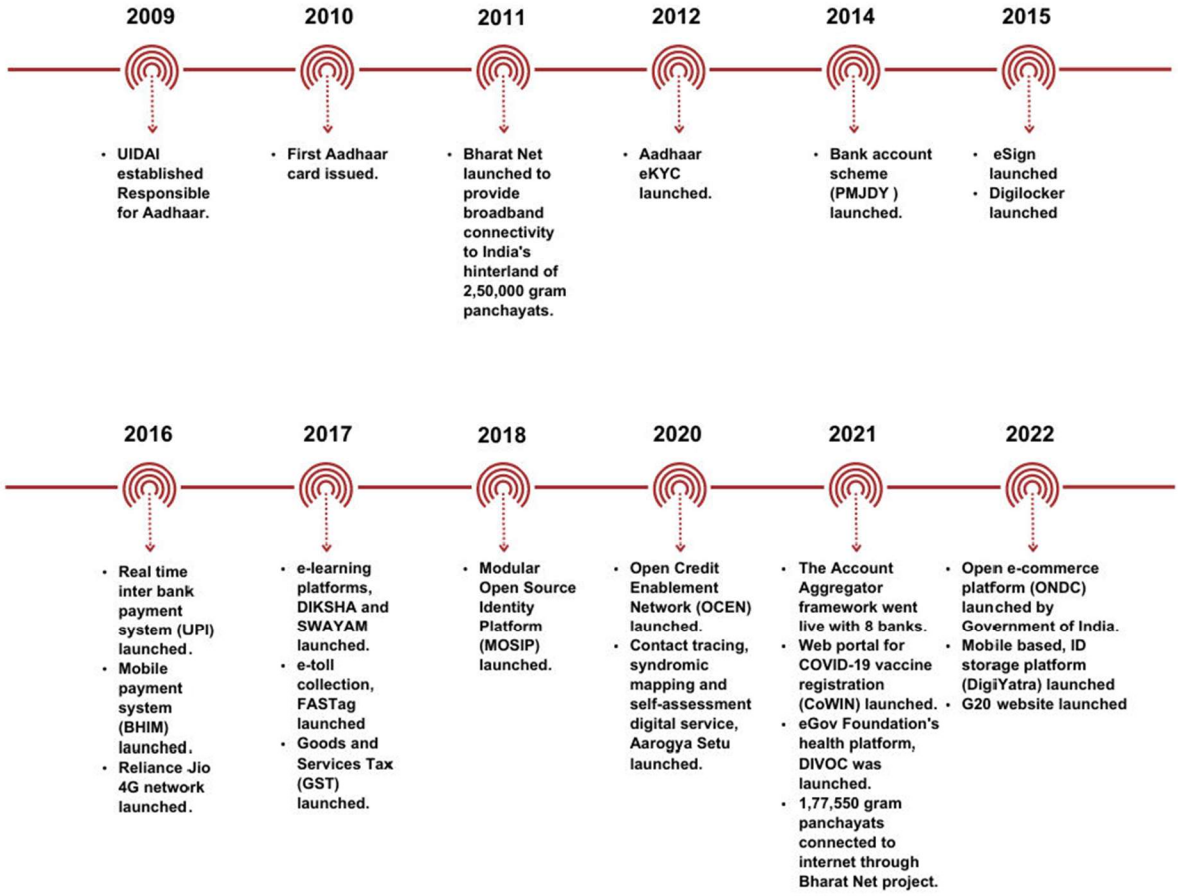
Exhibit 1b Principles Underlying Digital India Initiative

Ecosystem Principles	Ecosystem Thinking	Design all digital initiatives as ecosystems, NOT as systems.
	Building Block approach	Design ecosystems in terms of minimal and reusable Building Blocks.
	Open API-based	Adopt 'Open-API by default'. Exceptions shall be justified.
	Open, Open and Open	Build on open source, publish on open source, conform to open standards. Exceptions shall be justified.
	National Portability	Design digital systems and platforms for portability across India.
	Participatory Design	Drive participatory design and end-user engagement at all stages.
	Innovation	Enable and promote innovation, and 'responsible' deployment of emerging technologies.
Architecture Principles	Federated Architecture	Adopt a Federated Architecture model for designing digital ecosystems – especially data and applications.
	Agile Architecture	Permit the architecture to evolve in an agile and iterative manner.
	Technology Independence:	Architect systems to be technology-independent.
Business Principles or Domain Principles	Value-driven	Provide value to the user. Plan to Define, Design and Deliver value.
	Integrated Services	Identify, design, and deliver integrated services that cut-across agency boundaries, to realize the goal of Connected Government.
	Outcome-driven	Benchmark outcomes with the best, build services around such outcomes. Work backwards to re-engineer the processes.
	Choice	Provide choice, by design, to the citizen.
	Universal Access	Ensure digital services are accessible to all.
Technology Principles	Cloud First	Cloud shall be the first choice in taking ICT infrastructure decisions.
	Mobile First	Design the delivery of all digital services through mobile by default.
	Data is an asset	Design data systems to create, support, maintain, and enhance value to the enterprise specifically, and to the ecosystem in general.
	Data sharing	Clear data-sharing policies that enable and regulate the sharing of data, in conformance with the applicable data protection regulations.
	Standards	Specify the existing technology and data standards applicable to the ecosystem and define methods to ensure compliance with the same.
	Privacy-by-Design	Privacy policy to conform to the principles of Privacy-by- Design.
	Security-by-Design	Cybersecurity policy to conform to the principles of Security- by-Design, Information Security Management System to conform to information security standards
Governance Principles	Governance Systems	Accountable, transparent, and effective institutional structures for Architecture, IT, Data and Security Governance.
	Capabilities	Build capacities and capabilities in individuals and organizations.
	Business Model	Create a policy environment that supports financially viable and sustainable business models based on value-for-money.

Leverage the legacy	Leverage the existing systems after assessing the degree of their conformance to these principles and enhancing them appropriately for strict conformance.
Implementation Strategy	Think Big, Start Small, Scale Fast.
Enable rather than build	Enable the ecosystem to design, develop, deliver. Build by exception.

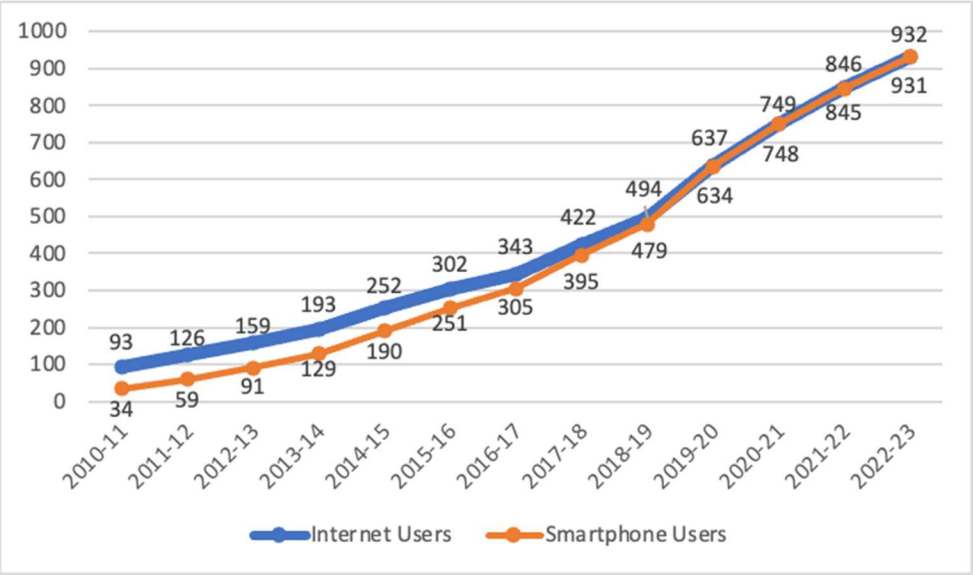
Source: Adapted by case writers from "InDEA 2.0 - India Digital Ecosystem Architecture 2.0," Ministry of Electronics & IT (January 2022), p.11-15, accessed May 2023.

Exhibit 1c Select Timeline of Building Blocks



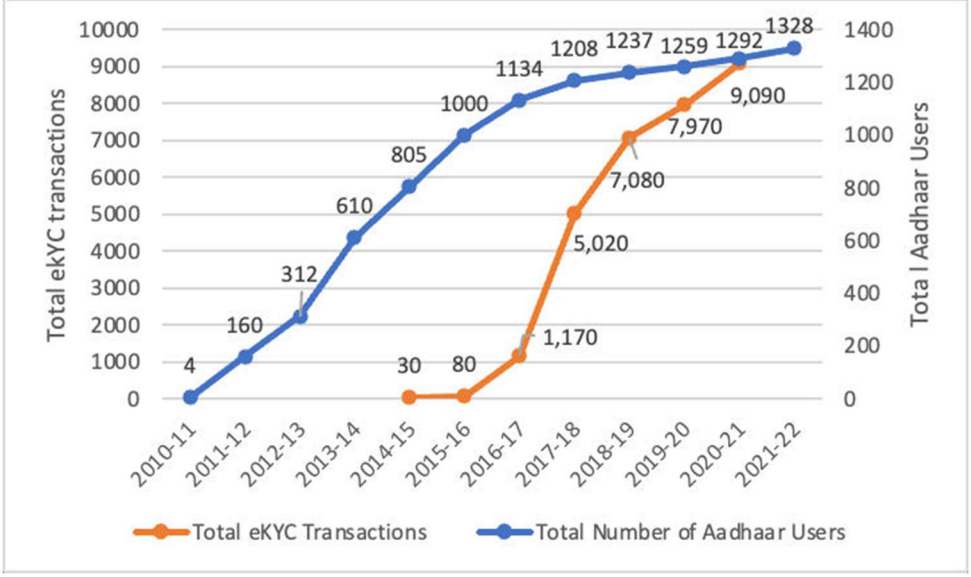
Source: Compiled by casewriters from India Stack, "About," https://indiastack.org/index.html, accessed May 2023.; Pradhan Mantri Jan Dhan Yojana (PMJDY), "About," https://pmjdy.gov.in/about, accessed May 2023.; BHIM UPI, "About Us," https://www.bhimupi.org.in/who-we-are, May 2023.; Bharat Bill Pay System, "About Us," https://www.bharatbillpay.com/about-us, accessed May 2023.; SWAYAM, "About Us," https://swayam.gov.in/about, accessed May 2023.; DIKSHA, "About Us," https://www.diksha.gov.in/about/, accessed May 2023.; Goods and Services Tax, "About Us," https://www.gst.gov.in/about/gst/history, accessed May 2023.; National Portal of India, "DigiYatra," https://www.india.gov.in/spotlight/digi-yatra-new-digital-experience-air-travellers, accessed May 2023.; DrishtiIas, "Bharat Net," https://www.drishtiias.com/daily-updates/daily-news-analysis/bharatnet-project, accessed May 2023; DIVOC by eGov Foundation, "About," https://divoc.egov.org.in/, accessed May 2023 and Suraksha P, "A foundational ID system to give identity to millions across the globe," The Economic Times, September 4, 2022, https://economictimes.indiatimes.com/tech/information-tech/a-foundational-id-system-to-give-identity-to-millions-across-the-globe/articleshow/93969853.cms?from=mdr, accessed May 2023. "Mukesh Ambani unleashes Reliance Jio, from September 5 to December 31, it will be free for all," The Economic Times, September 1, 2016, https://economictimes.indiatimes.com/news/company/corporate-trends/mukesh-ambani-unleashes-reliance-jio-from-september-5-to-december-31-it-will-be-free-for-all/articleshow/53957248.cms, accessed June 2023; Anuradha Mishra, "BharatNet scheme; creating a connectivity highway for all Gram Panchayats of India," News on Air, August 4, 2022, https://newsonair.com/2022/08/04/bharatnet-scheme-creating-a-connectivity-highway-for-all-gram-panchayats-of-india/, accessed June 2023; "Progress of National Broadband Mission," Ministry of Communications press release, July 22, 2022, on PIB website, https://pib.gov.in/PressReleaseIframePage.aspx?PRID=1843752, accessed June 2023.

Exhibit 2 Internet and Smartphone Users.



Source: Compiled by Case writers based on data from Statista, "Number of internet users in India from 2010 to 2020, with estimates until 2040," https://www-statista-com.ezp-prod1.hul.harvard.edu/statistics/255146/number-of-internet-users-in-india/, accessed March 2023; Statista, "Number of smartphone users in India in 2010 to 2020, with estimates until 2040," https://www-statista-com.ezp-prod1.hul.harvard.edu/statistics/467163/forecast-of-smartphone-users-in-india/, accessed March 2023.

Exhibit 3 Number of Aadhaar and e-KYC transactions.



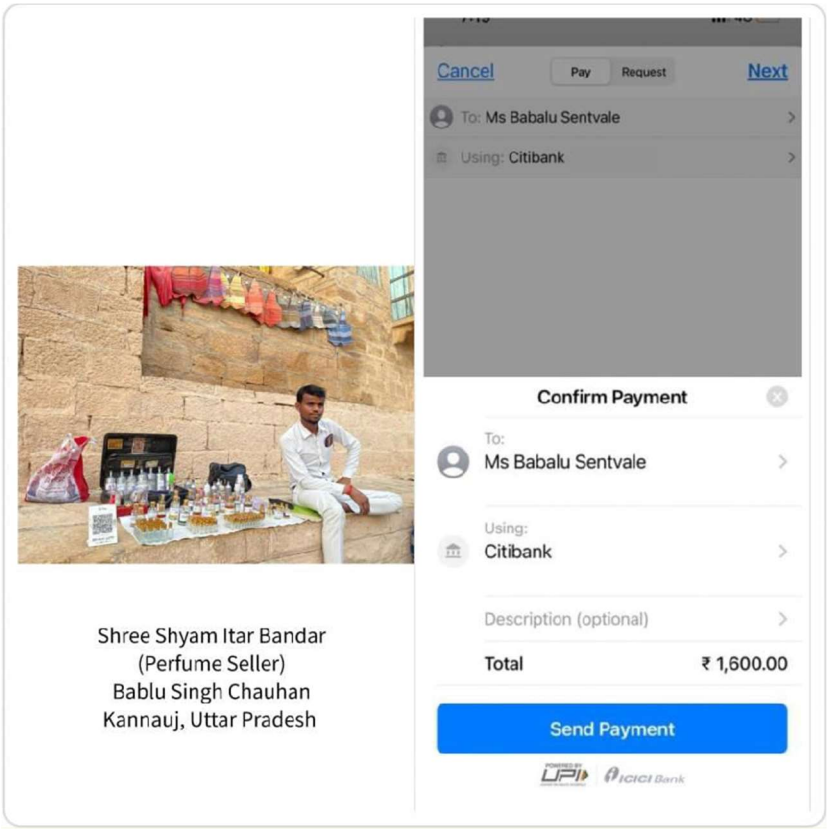
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Exhibit 4 The UPI payment process.

- Individuals and merchants had a UPI ID, i.e. a virtual payment address (Kabir@okhdfcbank), connected to a specific bank account. This ID did not reveal the bank account number or branch.[129]
- Payments could be initiated by either a "push" by the payer or a "pull" by the payee (which had to be accepted by the payer).
- One could choose to have up to four UPI IDs, one for each bank account, all connected to the same Aadhaar ID.
- To make a payment, a payer only needed a UPI ID or a mobile number linked to a UPI ID or a QR code. For example, street-side shops had a digital QR code that customers could scan to make payments.

Source Casewriters.

Exhibit 5 UPI at a small stall owner.



Source: Casewriters.

Exhibit 6a Statistics on the use of UPI.

Month	UPI users (Bn)	No. of Banks live on UPI	Volume (Mn)	Value ($Bn)
Jun'23	-	445	9,335.1	181
Mar'23	0.3	399	8685.3	172
Mar'22	0.26	314	5405.7	117
Mar'21	0.2	216	2731.7	61
Mar'20	0.14	148	1246.8	25
Mar'19	0.08	142	799.5	16
Mar'18	0.03	91	178.1	2.9
Mar'17	-	44	6.4	0.29
Dec'16	-	35	2.0	0.086
Nov'16	-	30	0.3	0.012
Apr'16	-	21	0.0	0.0

Source: Created by Case writers based on data from National Payments Corporation of India, "Product Statistics," https://www.npci.org.in/what-we-do/upi/product-statistics, accessed July 2023; 8.7 billion UPI transactions witnessed in March 2023, Nirmala Sitharaman, MoneyControl, April 2023, https://www.moneycontrol.com/news/business/economy/8-7-bn-upi-transanctions-witnessed-in-march-2023-nirmala-sitharaman-10419381.html, accessed July 2023.

Note: $1 = INR 82.21 as of March 31, 2023. Reference Rate Archives, RBI, https://www.rbi.org.in/scripts/ReferenceRateArchive.aspx, accessed July 2023.

Exhibit 6b UPI payment P2P and P2M.

	Total		P2P		P2M	
Month	**Volume (Mn)**	**Value ($ Bn)**	**Volume (Mn)**	**Value ($ Bn)**	**Volume (Mn)**	**Value ($Bn)**
Jun'23	9,335.1	179	3,971.43	137	5,363.63	43
Mar'23	8,685	172	3,859	132	4,826	39
Mar'22	5,406	117	3,201	95	2,205	22
Mar'21	2,732	61	1,516	51	1,216	10
Apr'20	1,000	18	550	16	449	3

Source: Adapted by case writers based on data from NCPI, "UPI Ecosystem," https://www.npci.org.in/what-we-do/upi/upi-ecosystem-statistics#innerTabThreeMar23, accessed April 2023.

Note: 1) P2P - Peer to Peer Payments and P2M - Peer to Merchant Payments

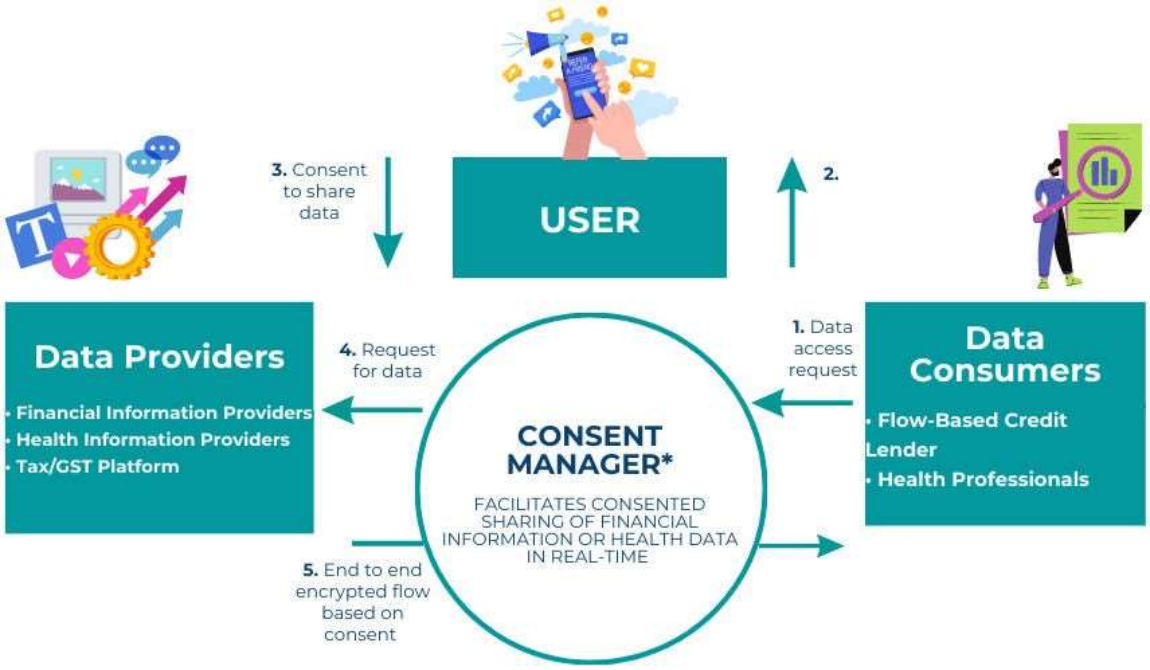
2) $1 = INR 82.21 as of March 31, 2023. Reference Rate Archives, RBI, https://www.rbi.org.in/scripts/ReferenceRateArchive.aspx, accessed July 2023.

Exhibit 7 Illustrative examples of DPI in EdTech and HealthTech.



Sources: Compiled by case writers from Digital Infrastructure for Knowledge Sharing, "About Diksha," https://diksha.gov.in/about/, accessed June 2023; Byju's, "DIKSHA - National Teachers Platform For School Education," https://byjus.com/current-affairs/diksha-digital-infrastructure-knowledge sharing/#:~:text=DIKSHA%20stands%20for%20Digital%20Infrastructure%20for%20Knowledge%20Sharing, accessed June 2023; Swayam, "About Swayam," https://swayam.gov.in/about, accessed June 2023; CoWin, "About," https://www.cowin.gov.in/, accessed June 2023; Ministry of Education, "SWAYAM Portal," https://pmevidya.education.gov.in/swayam-portal.html, accessed June 2023; National Payments Corporation of India, "Unified Payments Interface (UPI) Product Overview," https://www.npci.org.in/what-we-do/upi/product-overview, accessed June 2023; Open Network for Digital Commerce, "About," https://ondc.org/about-ondc/, accessed June 2023. Rahul Sanghi and Aaryaman Vir, "OCEN: A Conversation," Tigerfeathers, September 19, 2020, https://tigerfeathers.substack.com/p/ocen-a-conversation, accessed June 2023, and Bhashini, "About Bhashini," https://bhashini.gov.in/en/about, accessed July 2023.

Exhibit 8 Consent Manager



Source: Infographic created by Tanisha Murdeshwar, Research and Educational Coordinator at HBS India Research Center based on data from Data Empowerment and Protection Architecture, Anna Roy, p.4, https://www.niti.gov.in/sites/default/files/2020-09/DEPA-Book.pdf, accessed June 2023; and interview with Mr. Rahul Matthan, by Anjali Raina and Kairavi Dey in Mumbai, Maharashtra, via Zoom, June 2, 2023.

Note: Account Aggregator in the financial ecosystem.

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